Introduction to Knot Theory

Lecture 7: Links and Braids

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Definitions: Knots and Links

Knots and links are characterized by their homeomorphic properties within spheres.

Formal Definitions

- **Knot (K):** An embedding $K \subseteq S^n$ such that $K \cong S^p$.
- **Link (L):** An embedding $L \subseteq S^n$ such that $L \cong S^{p_1} \amalg \cdots \amalg S^{p_r}$ (a disjoint union of spheres).

- A link is a collection of one or more disjointly embedded circles (S^1).
- The ambient space is typically considered to be S^3 or $\mathbb{R}^3$.
- Note that there is only one type of knot of S^p in S^q whenever the codimension q - p is greater than two. How about links?

Example 1: $S^0 \cup S^0$ in S^1

- **Discrete Case:** Two points (S^0) embedded in a circle (S^1).
- Represents the simplest discrete link structure in lower dimensions.

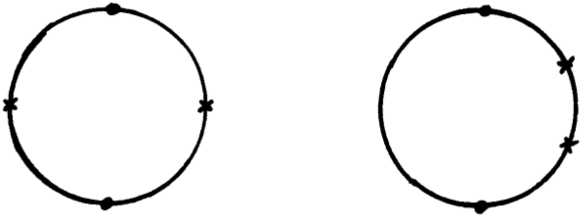

Figure: $S^0 \cup S^0$ in S^1

Example 2: $S^0 \cup S^1$ in S^2

- **Mixed Dimensions:** A pair of points (S^0) and a circle (S^1) within a 2-sphere (S^2).
- Illustrates the coexistence of different dimensional spheres within a manifold.

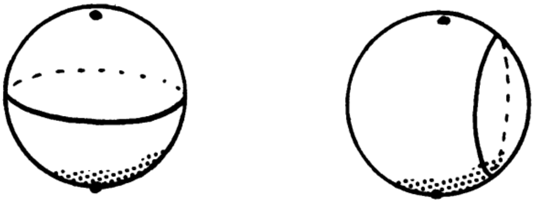

Figure: $S^0 \cup S^1$ in S^2

Example 3: $S^1 \cup S^1$ in S^3

- **Standard Case:** Two circles (S^1) embedded in a 3-sphere (S^3).
- The primary object of study in classical knot theory (e.g., Hopf link).

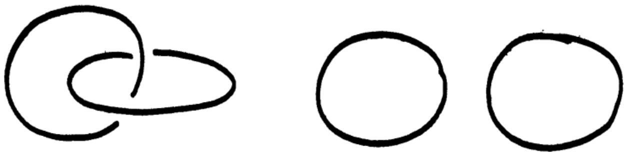

Figure: $S^1 \cup S^1$ in S^3

Example 4: Generalization to S^{p+q+1}

- **High-Dimensional Join:** A generalization representing multi-dimensional components.
- Constructed via the join of two spheres: $S^{p+q+1} \cong (S^p * S^q)$.
- Ambient space: $S^{p+q+1} = (\mathbb{R}^{p+1} \times \mathbb{R}^q) \cup \{\infty\}$.

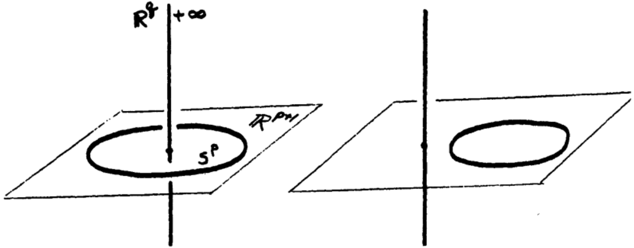

Figure: Conceptual Diagram of $S^p \cup S^q$ in S^{p+q+1}, which are not equivalent

Inequivalence of Links in S^{p+q+1}

To show that two links are inequivalent, we demonstrate that in the first case, S^p cannot be shrunk to a point in $S^{p+q+1} \setminus S^q$.

Topological Complement

The complement of S^q in the higher-dimensional sphere is given by:

$$S^{p+q+1} \setminus S^q \cong (\mathbb{R}^{p+1} \setminus \{a\}) \times \mathbb{R}^q$$

We define a **retraction** $r : S^{p+q+1} \setminus S^q \to S^p$ by the formula:

$$r(x, y) = \frac{x - a}{\|x - a\|}$$

Inequivalence of Links in S^{p+q+1}

Proof by Contradiction

Suppose there exists a homotopy $h_t : S^p \to S^{p+q+1} \setminus S^q$ from the inclusion to a constant map.

- Then $r \circ h_t : S^p \to S^p$ would be a homotopy between the **identity map** id_{S^p} and a **constant map**.
- This is impossible because the sphere S^p is not contractible ($\pi_p(S^p) \cong \mathbb{Z}$).

Conclusion: The first link is non-trivial, while the second is equivalent to the unlink.

Crossing Sign Convention

- Let's get back to the case of $\coprod_{i=1}^{n} S_i^1 \hookrightarrow S^3$
- The crossing signs are determined by the relative orientation of the strands using the **right-hand rule**.

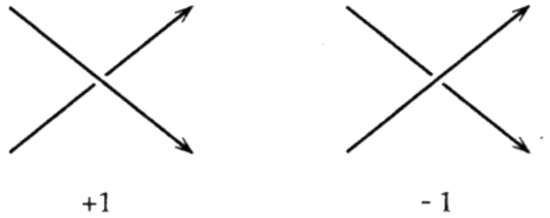

Figure: Crossing Signs: Positive (Right-handed, $+1$) and Negative (Left-handed, -1)

The Linking Number (lk)

The linking number is a numerical invariant that quantifies the net entanglement between two oriented components.

Mathematical Definition

For a link with two oriented components C_1 and C_2, the linking number is defined as:

$$lk(C_1, C_2) = \frac{1}{2} \sum_{p \in C_1 \cap C_2} \mathrm{sign}(p)$$

- It can be shown that the linking number is a **topological invariant**.
- Specifically, it remains invariant under **Reidemeister moves**.

Linking Number: Case Studies

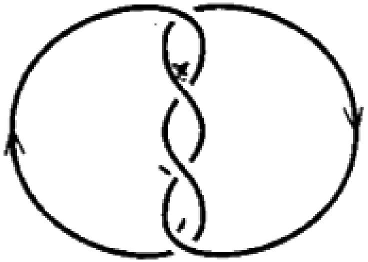

Figure: Link with $lk = 2$

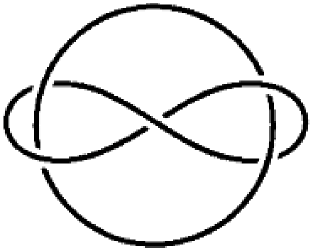

Figure: Whitehead Link ($lk = 0$)

- **Standard Link:** Components wrap around each other twice.
- **Whitehead Link:** Demonstrates that $lk = 0$ does not imply the link is trivial.

The Zero Paradox: Inquiry

Question

Does a **linking number of 0** imply that the link is trivial (unlinked)?

- **Intuition:** Zero winding suggests the components are independent.
- **Reality:** Link topology often involves complex, higher-order interactions that the linking number cannot detect.

The Zero Paradox: Borromean Rings

Counter-example: Borromean Rings

The Borromean Rings provide a definitive "No" to the paradox.

- Every pairwise linking number is 0.
- Despite this, the three-component system is globally inseparable.
- **Brunnian Property:** Removing any single component renders the remaining components trivial.

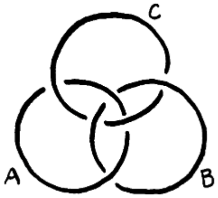

Figure: The Borromean Rings

Recall The Knot Group

Fundamental Group Definition

The **knot group** of a knot K is defined as the fundamental group of its complement:

$$G(K) = \pi_1(S^3 \setminus K)$$

- Encodes the topological embedding of the knot in space.
- This algebraic definition extends naturally to links L where $G(L) = \pi_1(S^3 \setminus L)$.

Wirtinger Presentation

Using the Wirtinger algorithm, we present the group via generators (arcs) and relations (crossings):

$$G(K) = \langle \text{arcs} \mid \text{crossing relations} \rangle$$

Example (Trefoil): $G(K) = \langle a, b \mid aba = bab \rangle$.

Why Borromean rings are inseparable?

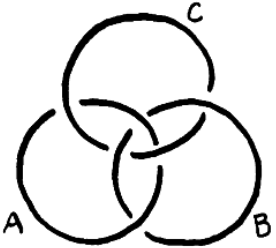

Figure: Borromean Rings

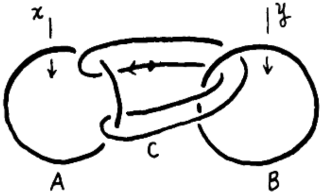

Figure: Borromean Rings with Generators

- Component C corresponds to the **commutator** $xyx^{-1}y^{-1}$ within the group $\pi_1(S^3 \setminus (A \cup B))$.
- Since $xyx^{-1}y^{-1} \neq 1$, the components are non-trivially linked.

A Magic Trick!

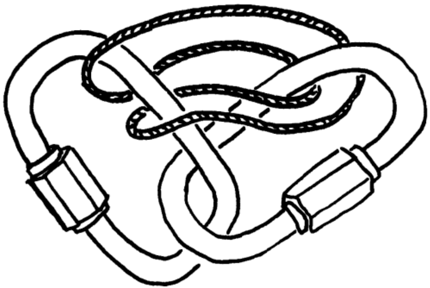

Figure: **Slipping off the central component:** In the Hopf complement, the commutator relation collapses, allowing the string to be removed.

Borromean

$$xyx^{-1}y^{-1} \neq 1$$

(Non-trivial interaction)

$\longrightarrow$

Hopf-like

$$xyx^{-1}y^{-1} = 1$$

(Commutative structure)

Definition of n-string Braid

Geometric Setting

Let $I^2 \times I$ be the unit cube in $\mathbb{R}^3$, and let n be a positive integer. We define two sets of n points on the top and bottom faces:

- **Top points:** $P_i = \left(\frac{i}{n+1}, \frac{1}{2}, 1\right)$ for $i = 1, \ldots, n$.
- **Bottom points:** $Q_i = \left(\frac{i}{n+1}, \frac{1}{2}, 0\right)$ for $i = 1, \ldots, n$.

Definition (n-string Braid)

An ***n*-string braid** b is a collection of n disjoint polygonal arcs $b_1, \ldots, b_n$ such that:

1. The set of endpoints is $\{P_1, \ldots, P_n, Q_1, \ldots, Q_n\}$.
2. Each arc is **monotonic** with respect to the z-coordinate.

Geometric Representation of Braids

Definition (n-string Braid)

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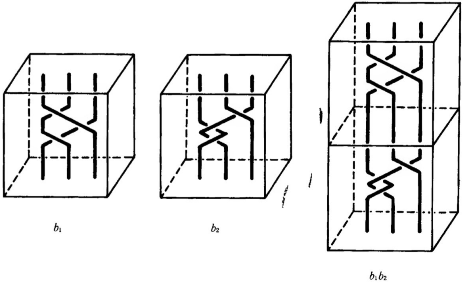


Figure: A geometric n-string braid. Note that strings only move downwards (monotonicity), preventing loops that go "upwards".

From Braids to Knots and Links: The Closure Property

From Braids to Links

Every braid corresponds to a particular knot or link through a geometric process called **closure**.

Geometric Construction:

- Pull the bottom bar of an n-string braid around.
- Glue it to the top bar, connecting corresponding endpoints.
- The resulting disjoint closed curves form a knot (1 component) or a link (>1 components).

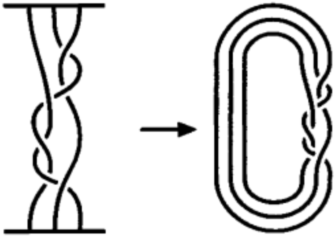


Figure: The closure of a braid.

From Knots and Links to Braids: Alexander's Theorem (1923)

The Fundamental Theorem

What knots and links can be represented as closed braids?

Amazingly enough, they all can.

- **Theorem:** Every knot or link is equivalent to a closed braid.

- **Historical Note:** First proved by **J. W. Alexander** in 1923.
- **Significance:** This bridge between Knot Theory and Braid Theory allows us to study knots using the algebraic structure of the Braid Group B_n.

Proof Strategy: Using Bridge Decomposition

To prove that any link L is a closed braid, we utilize the concept of **bridges**.

Step 1: Partitioning the Strands

- Orient each component of the link L.
- Choose points $P_1, \ldots, P_n$ along the strands between overcrossing and undercrossing.
- P_1 is chosen immediately after an **undercrossing**.

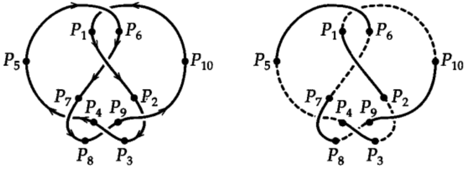


Figure: Labeled points P_i and bridge structure.

Step 2: Alternating Structure

- P_{2i-1} to P_{2i}: **Upper bridge** (above the projection plane).
- P_{2i} to P_{2i+1}: **Lower bridge** (beneath the projection plane).

This decomposition ensures the knot can be viewed as segments alternating across the projection plane.

Interlude: The Bridge Number $br(K)$

Definition

The **bridge number** of a knot K, denoted by $br(K)$, is the minimum number of bridges (upper bridge, over-strands) required among all possible diagrams of K.

Key Characteristics:

- $br(K) = 1$ if and only if K is the **Unknot**.
- For the **Trefoil Knot**, $br(K) = 2$.
- Geometrically, it corresponds to the minimum number of local maxima in a height function.
- Alexander's proof essentially converts these n bridges into n braid strands.

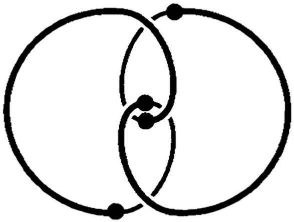


Figure: Trefoil knot with $br(K) = 2$.

Step 3: Aligning the Lower Bridges

Isotopy under the Projection Plane

We rearrange the projection so that the n strands **beneath** the projection plane are lined up systematically.

The Rearrangement:

- Even-numbered points ($P_2, P_4, \dots$) are aligned next to one another.
- Lower bridges are simplified into parallel arcs (as in Figure 5.38).
- **Safety:** Since these arcs do not intersect, this isotopy is always possible without cutting.

Note: While the lower side becomes simpler, the upper bridges may become "messier" but remain disjoint.

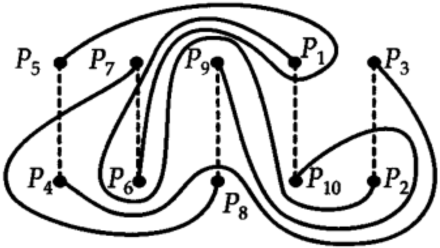


Figure: Lined-up lower bridges.

Step 4: Bridge Work – Normalizing Crossings over Axis A

The Parity Problem

- Upper bridges start North of axis A and end South of A.
- To form a braid, we need **exactly one** crossing per bridge.

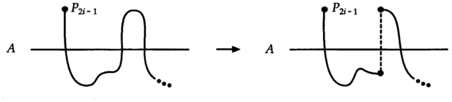


Figure: Figure 5.39: Reducing crossings by creating a new lower bridge.

Reduction Algorithm

1. Identify a bridge with > 1 crossings.
2. At the 2nd crossing point, **push the strand beneath** the projection plane.
3. Repeat until all upper bridges are monotonic.

After this process, every strand travels from North to South across A exactly once.

Step 5: Final Construction of the Closed Braid

3D Spiral Configuration

- **Upper Bridges ($P_{2i-1} \to P_{2i}$):**
  Rise above axis A at the crossing point, then descend back to the projection plane.
- **Lower Bridges ($P_{2i} \to P_{2i+1}$):**
  Dip beneath axis A at the crossing point, then rise back to the projection plane.

Conclusion of the Proof

- Every strand now flows **monotonically** around the central axis A without backtracking.
- This configuration satisfies the geometric definition of a **closed braid**.
- **Alexander's Theorem:** Every knot or link can be represented as a closed braid.

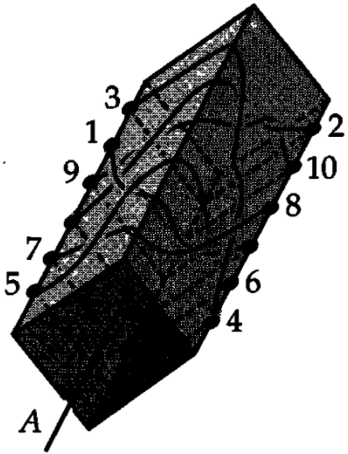


Figure: The closed braid around axis A.

Transition: Why Braid Groups?

1. Limitation of Geometric Representation

Alexander's Theorem ensures a braid exists, but the resulting 3D geometry is often too complex to analyze visually. We need a **rigorous algebraic language** to simplify and calculate these structures.

2. Ultimate Goal: Classification of Links

The algebraic study of braids allows us to:

- **Distinguish Links:** Systematically determine if two different-looking links are equivalent.
- **Construct Invariants:** Build powerful tools like the **Jones Polynomial** through braid algebra.

The Braid Group B_n

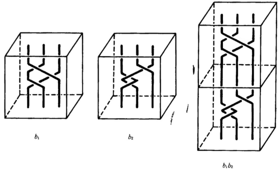
Equivalence and Multiplication

- **Equivalence:** Two braids are equivalent ($b \approx b'$) if there is an ambient isotopy fixing the boundary $\partial(I^2 \times I)$.
- **Multiplication ($b \cdot b'$):** Defined by "stacking" braid b on top of b' and rescaling the z-coordinate.

Group Structure

The set of all equivalence classes of n-string braids forms the **braid group**, denoted by B_n.

- **Identity:** n vertical straight lines.
- **Inverse:** The mirror image of the braid reflecting across the $z = 1/2$ plane.



The Braid Group B_n

Definition

The **Braid Group** B_n is defined by generators $\sigma_1, \dots, \sigma_{n-1}$ with:

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \mathcal{R} \rangle$$

where the relations $\mathcal{R}$ are:

$$\sigma_i \sigma_j = \sigma_j \sigma_i \qquad \text{for } |i-j| \geq 2 \tag{1}$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \qquad \text{for } 1 \leq i < n-1 \tag{2}$$

- **Generators:** σ_i is the crossing of i-th and $(i+1)$-th strands.
- **Relations:** Braids are equivalent under ambient isotopy.

The Braid Group B_n

$$\langle \sigma_1, \ldots, \sigma_{n-1} \mid \sigma_i\sigma_k = \sigma_k\sigma_i(|i-k| \geq 2),$$
$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}(i \leq n-2)\rangle$$

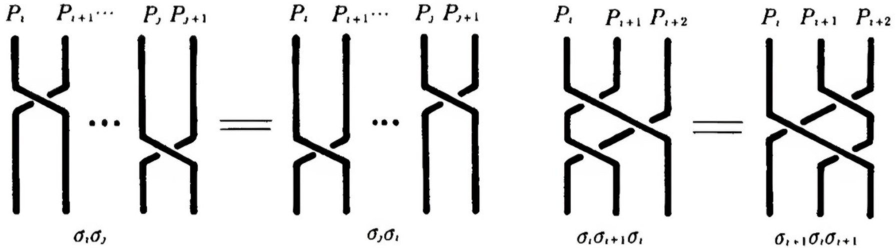


Figure: braid group presentation

Markov's Theorem and Braid Relations

Markov's Theorem

Two braids β, β' represent the same link L (i.e., $\hat{\beta} \approx \hat{\beta}'$) if and only if they are related by a finite sequence of moves. This equivalence is governed by the algebraic relations in B_n and the Markov moves.

Defining Relations of Braid Group B_n

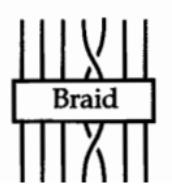
For any generators σ_i, σ_j of the braid group:

1. $\sigma_i \sigma_j = \sigma_j \sigma_i \quad$ for $|i - j| \geq 2$
2. $\sigma_i \sigma_i^{-1} = e$ (Reidemeister move 2)
3. $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ (Reidemeister move 3)

Markov Moves: Closure Equivalence

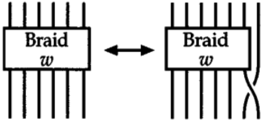
Type I: Conjugation

For $\beta, \alpha \in B_n$, the move $\beta \sim \alpha\beta\alpha^{-1}$ preserves the closure. This corresponds to sliding a braid segment around the axis.



Type II: Stabilization

For $\beta \in B_n$ and $\sigma_n^{\pm 1} \in B_{n+1}$, the move $\beta \sim \beta\sigma_n^{\pm 1}$ preserves the closure. This increases or decreases the strand count.



$w\sigma_n$

Markov's Theorem: Links are the same $\iff$ Braids are Markov equivalent.

References

Core Literature

- Rolfsen, D. (1976). *Knots and Links*. Wilmington, DE: Publish or Perish.
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- Kawauchi, A. (1996). A Survey of Knot Theory. Birkhäuser Basel.

Image Credits

- Diagrams adapted from Rolfsen (1976), Adams (1994) and Kawauchi(1996).